

Midterm (practice version - NOT GRADED), Math 170S, Winter 2018
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Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems.
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- Books, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. For this problem only: please provide short answers (or short explanations when appropriate).

- (a) (5 points) Let $X_1, X_2, \dots, X_{20}$ are samples taken from normal $N(\mu_X, 2)$ distribution ($2 = \sigma^2$), $Y_1, Y_2, \dots, Y_{40}$ are samples taken from normal $N(\mu_Y, 1)$ distribution, $Z_1, Z_2, \dots, Z_{45}$ are taken from normal $N(\mu_Z, 5)$ distribution. All X_i 's, Y_i 's and Z_i 's are mutually independent. What is the distribution of a random variable $\bar{X} - 2\bar{Y} + 3\bar{Z}$?

Solution

For any random variable $Q \sim N(\mu, \sigma^2)$, $\bar{Q} \sim N(\mu, \frac{\sigma^2}{n})$. So, using properties of expectation and variance of the linear combination of gaussian random variables,

$$\bar{X} - 2\bar{Y} + 3\bar{Z} \sim N(\mu_X - 2\mu_Y + 3\mu_Z, 2/20 + 2^2 \cdot 1/40 + 3^2 \cdot 5/45).$$

- (b) (5 points) Find 90% confidence interval for a parameter $\mu_X - 2\mu_Y + 3\mu_Z$.

Solution

$$\mathbb{P}(-z_{0.05} \leq \frac{\bar{X} - 2\bar{Y} + 3\bar{Z} - (\mu_X - 2\mu_Y + 3\mu_Z)}{\sqrt{2/20 + 2^2 \cdot 1/40 + 3^2 \cdot 5/45}} \leq z_{0.05}) = 0.9.$$

So, 90% confidence interval for $\mu_X - 2\mu_Y + 3\mu_Z$ is

$$[\bar{X} - 2\bar{Y} + 3\bar{Z} - z_{0.05}\sqrt{1.2}; \bar{X} - 2\bar{Y} + 3\bar{Z} + z_{0.05}\sqrt{1.2}]$$

2. (10 points) We have 2 boxes, each containing 3 balls. Box number 1 contains one black and two white balls; box number 2 contains two black and one white ball.

Our friend chooses one of the boxes at random, probability of choosing box number 1 is p . Then he takes one ball from a chosen box (each of three balls can be taken chosen equally likely), and it turns out to be white.

We are going to find MAP estimate for the parameter $\theta = 1$ or 2 to conclude which box was chosen (θ is the number of the box). For what values of p we can conclude that box number 1 was chosen? For what values of p we can conclude that box number 2 was chosen?

Solution

Prior density (PMF) of choosing each box is $P(Box = 1) = p$, $P(Box = 2) = 1 - p$ (Box is the name of our parameter, it can take two values, 1 or 2, given the value of the Box we know conditional probabilities to draw a white and a black ball). Bayes theorem gives:

$$\begin{aligned} \mathbb{P}(Box = 1|Ball = white) &= \\ \frac{\mathbb{P}(Ball = white|Box = 1) \cdot \mathbb{P}(Box = 1)}{\mathbb{P}(Ball = white|Box = 2) \cdot \mathbb{P}(Box = 2) + \mathbb{P}(Ball = white|Box = 1) \cdot \mathbb{P}(Box = 1)} &= \\ \frac{2p/3}{2p/3 + (1-p)/3} &= \frac{2p}{1+p}. \end{aligned}$$

Then,

$$\mathbb{P}(Box = 2|Ball = white) = 1 - \frac{2p}{1+p} = \frac{1-p}{1+p}.$$

MAP estimate for the parameter Box is the point of maximum of the posterior density, which takes only 2 values: $\frac{2p}{1+p}$ and $\frac{1-p}{1+p}$. So, MAP estimate gives $Box = 1$ if

$$\frac{2p}{1+p} > \frac{1-p}{1+p},$$

that is, $2p > 1 - p$, $p > 1/3$. If $p < 1/3$, we would conclude (based on the MAP estimate) that box 2 was chosen. When $p = 1/3$ we cannot decide since posterior density is constant (and have several points of argmax).

3. Consider a sequence of independent coin tosses, and let θ be the probability of heads at each toss.

- (a) (4 points) Let k be a fixed number, and N is the number of tosses until k -th head occurs. Find maximum likelihood estimator of θ (depending of N).

Solution

Maximum likelihood function is just a density function, since only one sequence of experiments was performed). Density function is discrete,

$$\begin{aligned}\mathbb{P}(X = N) &= \mathbb{P}(\text{ exactly } k \text{ successes in } N \text{ tosses, and the last one is success }) \\ &= \binom{N-1}{k-1} \theta^k (1-\theta)^{N-k}.\end{aligned}$$

if $N \geq k$ and 0 otherwise.

To find MLE, take the derivative:

$$\text{const} \cdot [k\theta^{k-1}(1-\theta)^{N-k} - (N-k)\theta^k(1-\theta)^{N-k-1}] = 0,$$

so $k(1-\theta) - (N-k)\theta = 0$, so $\hat{\theta} = k/N$.

- (b) (4 points) Now let n be a fixed number, and K is the number of heads in n tosses. Find maximum likelihood estimator of θ (depending on K).

Solution

In this case density function is just binomial: probability that there are K successes in n tosses with success probability θ is

$$\mathbb{P}(X = K) = \binom{n}{K} \theta^K (1-\theta)^{n-K}.$$

Computing derivative and setting it to zero, we get exactly the same answer as in part a):

$$\text{const} \cdot [K\theta^{K-1}(1-\theta)^{n-K} - (n-K)\theta^K(1-\theta)^{n-K-1}] = 0,$$

so $K(1-\theta) - (n-K)\theta = 0$, so $\hat{\theta} = K/n$.

- (c) (2 points) Is the estimator from part (b) biased?

Solution

It is unbiased, since K is binomial $\mathbb{E}K = n\theta$, so

$$\mathbb{E}(\hat{\theta}) = \mathbb{E}(K/n) = \mathbb{E}(K)/n = n\theta/n = \theta.$$

4. (a) (4 points) Let $X_1, X_2, \dots, X_n$ are samples taken from the uniform distribution on $[\theta, 1]$ ($\theta < 1$). Show that $X_{(1)}$ is a sufficient statistic for θ .

Solution

Joint density of $X_1, \dots, X_n$ is

$$f(x_1, \dots, x_n; \theta) = \prod_{i=1}^n \frac{1}{1-\theta} I_{\{x_i \in [\theta, 1]\}} = \frac{1}{(1-\theta)^n} I_{\{\theta \leq x_1, \dots, x_n \leq 1\}} = \frac{I_{\{x_{(1)} \geq \theta\}}}{(1-\theta)^n} \cdot I_{\{\max_i x_i \leq 1\}}.$$

By factorization theorem, $u(X_1, \dots, X_n) = X_{(1)}$ is a sufficient statistic: $\varphi(u(x_1, \dots, x_n), \theta) = \frac{1}{(1-\theta)^n} I_{\{u(x_1, \dots, x_n) \geq \theta\}}$ and $h(x_1, \dots, x_n) = I_{\{\max_i x_i \leq 1\}}$ is independent from θ .

- (b) (6 points) Let $X_1, X_2, \dots, X_n$ are samples taken from the uniform distribution on $[\theta, \theta + 1]$. What is the length of a confidence interval with confidence coefficient $1 - \alpha$ for the estimate $X_{(1)}$ for parameter θ ?

Solution

It makes sense to consider one-sided confidence interval, since $X_{(1)} \geq \theta$ always (as $X_i \sim \text{Unif}[\theta, 1]$). So the goal is to find maximum length of the interval $[X_{(1)}, X_{(1)} - c_\alpha]$, so that true value of parameter θ is inside with the probability $1 - \alpha$:

$$1 - \alpha = \mathbb{P}\{X_{(1)} - c_\alpha \leq \theta\} = 1 - \mathbb{P}\{X_{(1)} - c_\alpha > \theta\} = 1 - \mathbb{P}\{X_1, \dots, X_n > c_\alpha + \theta\} =$$

from independence of the samples

$$1 - (\mathbb{P}\{X_1 > c_\alpha\})^n = 1 - (\mathbb{P}\{X_1 \in [c_\alpha + \theta, \theta + 1]\})^n = 1 - (1 - c_\alpha)^n.$$

So, $(1 - c_\alpha)^n = \alpha$ and the length of the interval $c_\alpha = 1 - \alpha^{1/n}$.

5. (10 points) Let $X_1, X_2, \dots, X_n$ be samples taken from some distribution with unknown density function. However, you know that $\mathbb{E}X = \alpha/\beta$ and $\text{Var } X = \alpha/\beta^2$, where α and β are two parameters of the density function (and there is no third parameter). What method could we use to find good estimates for α and β ? What are these estimates?

Solution:

Since we do not know density function, we cannot find maximum likelihood estimate, so let us find method of moments estimate. There are 2 parameters, so we need to find 2 moments.

First moment:

$$\mu_1 = \overline{X} = \tilde{\alpha}/\tilde{\beta}$$

Second moment:

$$\mu_2 = \frac{1}{n} \sum X_i^2 = \mathbb{E}(X^2) = (\mathbb{E}X)^2 + \text{Var } X = \tilde{\alpha}^2/\tilde{\beta}^2 + \tilde{\alpha}/\tilde{\beta}^2.$$

Solving system of equations for $\tilde{\alpha}$ and $\tilde{\beta}$, we get

$$\mu_2 - \mu_1^2 = \tilde{\alpha}/\tilde{\beta}^2$$

$$\frac{\mu_1^2}{\mu_2 - \mu_1^2} = \tilde{\alpha}$$

$$\frac{\mu_1^2}{\mu_2 - \mu_1^2} \frac{1}{\mu_1} = \frac{\mu_1}{\mu_2 - \mu_1^2} = \tilde{\beta}$$

Answer:

$$\tilde{\alpha} = \frac{\overline{X}^2}{\frac{1}{n} \sum X_i^2 - \overline{X}^2};$$

$$\tilde{\beta} = \frac{\overline{X}}{\frac{1}{n} \sum X_i^2 - \overline{X}^2},$$

where

$$\overline{X} = \frac{1}{n} \sum X_i.$$